

Timber World Trading Platform

System Specification

Architecture, deal lifecycle, roles and access

Prepared for the development team

Version 1.0 — for build

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This document defines the base system to be built now. Section 1.3 lists what is intentionally out of scope for this version.

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1. Purpose and scope

1.1 What this system is

The system is a trading platform for moving goods through a chain of independent companies. The business operates as an asset-light intermediary — buying from suppliers and selling to buyers, sometimes through several linked companies. Every transaction in the system is a deal between exactly two parties: one selling and one buying.

Goods typically pass through a chain of such deals. For example: the producer sells to the Latvian trading company, which sells to the UK trading company, which sells to the final customer. The same physical product may be identical at every step, but each handover is its own separate deal with its own price, terms and documents.

The platform has a single owner and operator — the super administrator — who can see the entire chain. Every other user (salesperson, buyer, producer, customer, accounting, warehouse) sees only the part of the business that concerns their own job. Keeping each party's view as narrow as possible is a core design goal, not an afterthought: it is what allows the business to sit in the middle of a multi-company chain without exposing suppliers to customers, or prices to parties who should not see them.

The platform is also built from the start to work with AI assistance for handling documents: a user can drop in an email, a PDF or a note, and the system extracts and structures the information into the deal's records and documents.

1.2 What we are building now

This specification covers the base of the system — the structural core on which everything else will later be built:

- The deal-and-chain structure and the spine that links them (Section 2).
- The identifiers and coding scheme (Section 3).
- The data model and entities (Section 4).
- The product catalogue and specification fields (Section 5).
- The deal lifecycle and stages (Section 6).
- The documents that live on each deal (Section 8).
- The user groups and the access model — who can see and do what (Section 9).
- AI-assisted document handling, inbound and outbound (Section 10).

The aim is a system that can be used immediately: users can create deals, enter specifications, generate and view documents, and confirm them — with the chain correctly linked behind the scenes.

1.3 What we are not building now (noted for the future)

The following are recognised as future directions and are deliberately out of scope for this version. They are recorded so the data model can accommodate them later, but they are not to be built now:

- A full CRM layer (task management, activity tracking, communication logs).
- Any rigid, enforced workflow. The system offers capabilities at each stage; the operator chooses what to do. Steps are not mandatory.
- An AI agent that resolves cancellations and exceptions automatically.
- Multi-country or multi-region automation beyond the basic structure.
- Sending transport requests to many carriers and comparing their replies (multi-carrier quoting).
- Automatic margin rules. For now, margin approval is manual (the owner).
- Rich reporting and dashboards beyond the basic per-deal status and the owner's overview.

1.4 Guiding principles

- **The system is the hub.** Parties do not communicate with each other directly through the system. Each party deals only with the system; the owner (and, in future, an AI agent) handles anything that needs coordinating across parties. This keeps a chain that may one day span several countries from becoming unmanageable.
- **Configurability lives in the application, not in code.** User groups, their rights, the confirmation gates, the product catalogue and the counterparty records are all created and changed by the administrator through the system's settings — never by reprogramming.
- **Minimum visibility.** Every user sees the least that lets them do their job. Access is denied by default and granted explicitly.

2. Core concepts (the model)

This section explains how the system works conceptually. The data fields in later sections only make sense on top of this model, so it should be read first.

2.1 Deals

A deal is a single transaction between exactly two parties — one seller and one buyer. It is self-contained: it has its own specification (what is being traded), its own agreed price, its own payment terms, its own delivery terms (Incoterms), and its own set of documents. A deal knows about its two parties and nothing else. There is no concept of “buying” inside a selling deal, or the reverse; a deal is purely the agreement between its own seller and buyer.

2.2 Chains

Goods usually move through several companies before reaching the final customer, and each handover is its own separate deal. A typical chain: the producer sells to the Latvian trading company (deal 1); the Latvian company sells to the UK trading company (deal 2); the UK company sells to the final customer (deal 3). Each deal is a distinct legal transaction with its own price and terms. These linked deals form a chain.

2.3 The spine — the thread that links the chain

Each deal is a closed, local link; it does not know about the other links. What connects them is a separate object called the spine. The spine represents the underlying batch of goods as a single, traceable identity that runs through the whole chain. It is the one and only place where the deals are connected to each other; the connection does not live inside any deal.

The spine's identity is born early — at the moment the goods are first defined (the first order/specification), before any physical goods exist — and it persists unchanged once the goods are physically made. The spine therefore has two life-stages of one identity:

- **Spec (planned):** the goods as described and intended, before they exist.
- **Lot (physical):** the same goods once produced, as a real, traceable batch.

The spine is the same thread before and after the goods exist; the physical lot inherits the identity created at the planning stage rather than getting a new one.

Splits and merges are handled by the spine through lineage in its identifier. If one order is filled by two factories, the spine splits into child spines (for example SP-042 becomes SP-042-A and SP-042-B), each traceable back to the parent. If parts come from different sources — some made to order, some from existing stock — several spines merge into one fulfilment. Splits and merges are expected to be uncommon but must be supported from day one.

A spine comes into being from either end of the chain:

- **Demand-driven:** a customer sale seeds a spine that flows upstream, creating the matching purchase(s).
- **Supply-driven:** existing stock or a production run is already a spine, waiting to be sold.

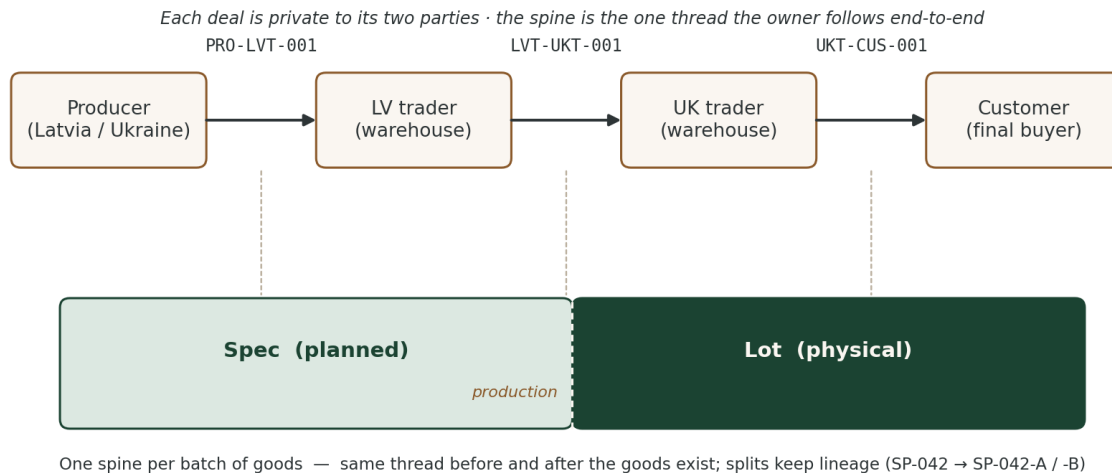


Figure 1. A chain of separate, private deals, all linked by one spine that runs from planned spec to physical lot.

2.4 Specification, order and lot

Three related but distinct ideas, which must be kept separate in the system:

- **Product definition / specification:** what the goods are — species, type, finish, certificate, dimensions, pieces, volume. This is the same product no matter how many times it changes hands.
- **Order:** the product definition plus one deal's commercial terms — its price, payment and delivery terms. The order differs per deal, because each leg has different terms.
- **Lot:** the spine once the goods physically exist.

So two deals in a chain can carry the same product definition (the same goods) while being different orders (different prices and terms). The shared product definition belongs to the spine, not to either deal.

2.5 Direction — sell deal and buy deal

From the operating company's standpoint, every deal is one of two kinds:

- **Sell deal:** the company is the seller; the deal faces a customer (downstream, toward the final buyer).
- **Buy deal:** the company is the buyer; the deal faces a supplier (upstream, toward the producer).

Within any single deal, the two parties are always simply seller and buyer — fixed by the deal, never ambiguous. The terms “sell deal” and “buy deal” describe a deal from the operating company's point of view, and are used throughout this document instead of “buyer side / seller side”, which are ambiguous.

Direction is relative to the viewer. The middle leg between the company's own two entities (the Latvian company selling to the UK company) is a single deal — one record, one set of documents — but the Latvian employee sees it as a sell deal and the UK employee sees it as a buy deal. This is expected and correct: direction is a property of the leg a person is standing on, not an absolute label baked into the deal.

2.6 How deals relate to the spine

- A deal is simple and local: its own two parties, its own terms, its own documents.
- The spine is the thread that runs through many deals and carries the connection between them.
- The owner follows the spine to see the whole chain; the parties never see the spine beyond their own deal.

This separation — simple local deals, one connecting spine — is the heart of the system.

3. Identifiers and codes

The system uses two primary identifiers, plus counterparty codes.

3.1 Deal code (directional)

Each deal has a unique directional code that shows who sells to whom, in the form SELLER-BUYER-NNN. Each company has a unique three-letter code; the deal code combines the seller's code, the buyer's code and a sequential number — for example TIM-DDC-001. The number runs to three digits (001 to 999); when it would roll over it can simply restart, since the volume per pair is low and would take years to exhaust. The directional code makes the goods' direction visible at a glance, and belongs to the deal (a deal is defined by who sells to whom).

3.2 Spine ID

Each batch of goods (each spine) has a unique spine identifier, created at the first order/specification and carried unchanged into the physical lot. Splits create child IDs that preserve lineage (SP-042 becomes SP-042-A, SP-042-B), so any child can always be traced back to its parent, whether the split was planned or happened during production.

3.3 Counterparty codes and other references

The three-letter company codes used in deal codes are drawn from the counterparty records (Section 4), so a code is defined once per company and reused. Orders, documents and physical shipments have their own references, but are always attached to a deal and/or a spine — they are not separate top-level numbering systems. Order content is the content of a deal; lot and package numbers identify the spine once it is physical.

4. Data model

The main objects in the system and what each one is:

Entity	What it is
Counterparty	A company record — a client or a supplier. Holds name, registration number, address, website, VAT number and a three-letter code. Deals reference counterparties rather than re-typing company details.
User	A person who logs in. Belongs to one or more groups, from which they inherit their rights.
Group	A named, editable set of rights (Section 9). Users inherit their access from their group.
Deal	A bilateral transaction between two counterparties (a seller and a buyer). Carries its order, its stage, its documents and its direction (sell or buy). References one spine.
Order / specification	The content of a deal: the product definition plus that deal's commercial terms (price, payment, delivery). Begins as a quotation and becomes the firm order specification.
Spine	The traceable identity of a batch of goods running through the chain. Has a spec (planned) phase and a lot (physical) phase, child spines for splits, and a rolled-up status. Referenced by every deal that handles the goods.
Lot / package	The spine once physical: the real, traceable goods, with package numbers.
Physical shipment	A movement of goods (a truck). Appears only when goods actually move, and may underlie one or several deals — in a back-to-back trade the goods can move once while ownership changes several times.
Document	A document attached to a deal: quotation, order specification, proforma invoice, sales contract, purchase order, packing list, CMR, invoices, and so on.
Product	A catalogue item with its specification fields (Section 5). Standard products carry an agreed price.

Relationships in brief:

- A deal references two counterparties (seller, buyer) and one spine.
- A spine is referenced by all the deals that handle its goods, forming the chain; it may have child spines (splits) and may merge several spines into one fulfilment.
- A user belongs to groups; groups grant rights; rights determine which deals and which fields the user can see and act on.
- Documents and physical shipments attach to deals and spines.

5. Products and specification fields

5.1 Product catalogue

The system holds a catalogue of products across the categories the business trades — for example edged boards, full-stave panels, tabletops, stair components, flooring, decking and firewood. Each product belongs to a category and carries a defined set of specification fields. New products and categories are added by the administrator in the application.

5.2 Specification fields (per item)

Each line item in a deal's specification carries the following standard fields; categories may extend the set:

- Package number
- Product name
- Wood species
- Type (for example full-stave or finger-jointed)
- Processing / finish (for example sanded, varnished)
- Humidity / moisture
- Certificate (for example FSC / PEFC, where applicable)
- Quality / grade
- Thickness, width, length
- Pieces (quantity)
- Volume (typically cubic metres)

5.3 Pricing of products

- **Standard products:** the catalogue holds agreed price levels; users select them and the price is applied automatically.
- **Non-standard products:** priced per deal — the selling price is entered on the sell deal, the buying price on the buy deal, and (for now) the owner approves the margin. Automatic minimum-margin rules are a future addition (Section 1.3).

6. Deal lifecycle (stages)

6.1 The stages

Every deal — sell deal or buy deal, anywhere in the chain — moves through the same five stages, plus a cancelled state. The stage is a milestone describing how far that one deal has progressed; it is not a sequence of mandatory steps.

- **Draft** — the deal is being prepared (specification, offer, contract, proforma) and agreed with its two parties.
- **Confirmed** — the deal is firm on both ends; preparation is finished. Nothing new is added here; it is the firm-and-waiting marker between agreeing the deal and making the goods.
- **Produced** — the goods for this deal are made.
- **Loaded** — the goods for this deal are loaded onto transport.
- **Delivered** — the goods arrive, are received and accepted; the deal closes.
- **Cancelled** — the deal is halted before delivery and flagged (see 6.4).

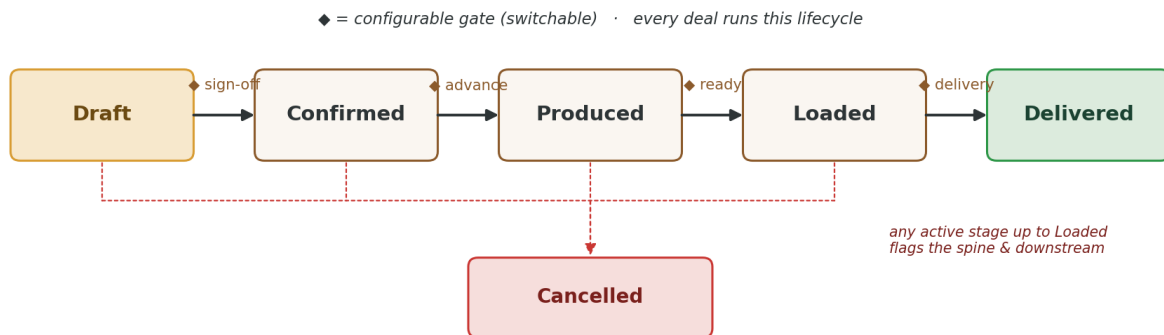


Figure 2. Every deal runs the same lifecycle. Each transition carries a configurable gate; cancellation is possible up to Loaded.

6.2 Two levels of status

Each deal carries its own stage. Each spine carries a rolled-up status derived from the stages of its deals. The deal stage runs the work; the spine status gives the owner the overview and flags a broken chain when any leg stalls. Ordinary users do not need the spine status; they see only a simple indication on their own deal (for example “in production”). The full spine overview is for the owner.

6.3 Confirmation gates (configurable)

Each transition from one stage to the next carries a gate — a set of required confirmations that must be satisfied before the deal advances. Gates are kept flexible because what is required differs from deal to deal: sometimes an advance payment, sometimes full payment, sometimes only a signature or a verbal agreement; sometimes a customer is not even on the system, sometimes a supplier confirms first. The gate is therefore an engine, configurable per deal-type and switchable on or off, assembled from a small set of building-blocks:

- **Party sign-off** — a named role confirms (for example salesperson, buyer, supplier or owner).
- **Condition met** — the system checks a fact (for example payment recorded, a document present, quality logged).
- **Acceptance** — the receiving party of a newly-created deal accepts it before it becomes active.

Any gate may be empty, in which case the deal advances on its own. Which confirmations apply to which transition is configuration entered by the administrator, not logic fixed in code. The specific confirmation rules — for example whether producer approval is required before a sale is confirmed — are deliberately left as configuration and are not fixed in this specification.

6.4 Cancellation and chain-break flagging

A deal can be cancelled from any active stage up to Loaded (not after Delivered). Cancelling a deal immediately flags the spine it belongs to and every deal downstream of that spine, so the owner and any party spanning the break can see it at once and decide what to do. This follows from the spine links and needs no separate mechanism. For now, the owner handles cancellations and exceptions; an automated resolution agent is a future addition (Section 1.3).

7. Activities by stage and direction

Every deal runs the same stages, but what a person does at each stage depends on the direction of the deal — facing the customer (sell deal) or facing the supplier (buy deal). The milestones are shared; the activities differ. Most of these activities are the CRM and communication layer; they are described here so the system is understood, but they are not built as enforced tasks in this version (Section 1.3). Transport ordering is included as an optional activity at the loading stage; it may sit on either side, or be handled by the customer or factory directly.

Stage	Sell deal (facing the customer)	Buy deal (facing the supplier)
Draft <i>preparing</i>	build the sales spec send the quotation negotiate terms with the customer prepare the sales contract & proforma secure the customer's agreement	build the purchase spec request a quote from the producer agree price, terms & timing with the factory prepare the purchase order secure the producer's agreement
Confirmed <i>firm</i>	deal locked as firm keep the customer informed of progress	deal locked as firm chase the factory on production progress
Produced <i>goods made</i>	confirm the goods are made and acceptable notify the customer	confirm production complete inspect / quality-check request photos & packing details verify against the spec prepare the packing list & labels
Loaded <i>goods loaded</i>	order transport for delivery (when it is your responsibility) confirm dispatch toward the customer send the shipment notice	order transport for the inbound leg (when it is your responsibility) confirm loading at origin obtain the transport documents (CMR)
Delivered <i>goods arrive</i>	customer receives & accepts close the deal	receive & unload into the warehouse check on arrival close the deal

Cancelled is identical on both directions: the deal is halted before delivery and flagged across the spine.

8. Documents

8.1 Documents belong to the deal

Because every deal passes through every stage, every document is available on every deal; the stage does not gate which documents exist. What the stage drives is the automation — what is sent or received, and when — which is the AI and automation layer (Section 10), not the document structure. So the document set simply belongs to the deal.

8.2 The document set

- **Quotation → order specification** — one document in two states: a non-binding quotation that becomes the firm order specification once accepted, issued as a PDF the counterparty receives.
- **Proforma invoice** — advance / payment request.
- **Sales contract** — for the sell side and new clients.
- **Purchase order** — the buy-side order to the supplier; belongs to the buy deal.
- **Packing list**.
- **CMR / transport documents**.
- **Invoices** (advance and balance) and payment records.

Documents are mostly shared across directions; a few are direction-specific (for example, packing labels prepared on the buy side at the Produced stage). These specific cases are few and attach to the relevant deal.

8.3 Outbound document pack

A deal supports a one-button send of the document pack a transport company needs — invoices, loading lists, packing details and addresses. See Section 10.

9. Roles and access

The access model is the central reason for the system: each party must see only what concerns its own job.

9.1 Two layers of access

- **Deal-level isolation:** a user sees only the deals they are a party to. This is what hides the chain — the middle leg between the company's own entities is a deal only the owner is on, so neither end of the chain can see it.
- **Field-level visibility:** within a deal a user can see, a field filter decides which fields are shown (for example, a logistics view shows transport fields but not pricing).

Both layers are driven by the user's group.

9.2 The group-and-rights model

Permissions attach to groups; users inherit from their group. The rights of a group are granular and editable in the application, so a new situation is handled by creating a group or changing a checkbox in settings — never by reprogramming. Rights are of three kinds:

- **Action rights** — what a user may do (create a sale, edit a spec, issue a document, confirm a stage, approve margin, cancel a deal, create a counterparty record).
- **Visibility rights** — what a user may see (sell side, buy side, pricing, supplier identity, spine status).
- **Scope** — which deals a right applies to (only mine, my company's, everything).

A group is simply a saved set of these. The system ships with sensible default groups, all editable:

Role	Sees	Can do
Super admin <i>the owner</i>	Every deal in every chain, including the hidden middle leg between your own entities — all fields, prices and supplier identities.	Everything: create, edit, confirm, approve margins, cancel; and configure groups, rights, gates, products and counterparties.
Salesperson <i>sell side</i>	Own sell deals only — spec, price, terms, documents, status — plus client records. Not upstream deals, supplier identity or upstream prices.	Create a sale, edit the spec, issue documents, confirm the sell deal, create and edit client records, send the transport pack on own deals.
Purchasing <i>buy side</i>	Own buy deals only — purchase spec, buy price, terms, supplier, documents, status — plus supplier records. Not sell-side prices or the customer.	Price the buy deal, place the purchase order, confirm the buy deal, create and edit supplier records, send the transport pack on own deals.
Client <i>customer login</i>	Their own deal only — their spec, their documents, status. Nothing about the	View, accept their order, upload their own purchase order.

Role	Sees	Can do
	goods' origin.	
Producer <i>factory login</i>	Only the production deal where they are the seller — production spec, terms, packing needs, status. Nothing onward in the chain.	Accept the order, confirm production, upload photos and packing lists.
Accounting	The financial fields of their entity's deals — proformas, invoices, payment status.	Issue and record invoices and payments.
Warehouse	Logistics fields of deals touching their warehouse — packing, loading, quantities, delivery. Not pricing.	Confirm loading, receipt and stock.

An outsourced sales agent is just a configurable salesperson, scoped to their own deals. Every row above is editable: a group is added, or a box ticked, in settings.

9.3 Counterparties (the address book)

Clients and suppliers are stored as counterparty records (Section 4), separate from the deals that reference them. Two roles manage their own side of the address book, and the visibility wall carries over to the records:

- **Salesperson** — creates and edits client records (name, registration, address, website, VAT, and so on) and selects them onto sell deals.
- **Purchasing** — creates and edits supplier and producer records (the same fields) and selects them onto buy deals.

The two address books are walled: the sales side does not see suppliers, and vice versa. The owner sees both. Whether purchasing may reuse a client record, or the lists stay fully separate, is an administrator setting.

9.4 Gates and groups are separate

A group says what a user is allowed to do; a gate says what must happen before a deal advances. They meet at confirmation time but are configured in different places and must not be conflated.

9.5 Logistics and customs — an outbound channel, not a role

Logistics and customs companies do not get logins; they would not use the system. They are an outbound document channel: a deal supports a one-button send of the transport-document pack to them, and their replies are read back into the deal's transport fields by the AI (Section 10).

They sit entirely outside the system. The future multi-carrier quote-and-compare feature is noted in Section 1.3; only the supporting fields are provided now.

10. Automation and AI

The system works with AI assistance from the start, for documents only — not for workflow:

- **Inbound ingestion:** a user drops in an email, a PDF or notes; the AI extracts the specification fields and commercial details and structures them into the deal. Users may also enter data directly from catalogue dropdowns. Both paths are supported.
- **Document generation:** documents are generated from the deal's data (quotation / order specification PDF, proforma, contract, purchase order, packing list, and so on).
- **Spawning the buy side:** when a sale needs sourcing, the system creates the matching buy deal on the same spine, stamped with the same spine ID at the moment of creation, so the sell and buy deals are linked automatically rather than reconnected by hand. A sale fulfilled from stock already held by the selling entity spawns nothing (that spine already exists); a sale fulfilled from stock in another of the company's warehouses spawns the buy deal as normal. The test is whether fulfilling the sale needs a leg from another entity.
- **Outbound to logistics:** a one-button send of the transport-document pack to a carrier or broker; their reply is read by the AI and used to update the deal's transport fields automatically.

Everything beyond this — automated cancellation handling, multi-carrier quote comparison, multi-country automation, automatic margin rules, richer reporting — is future direction (Section 1.3). The data model should carry the fields these will need, but the features are not built now.

11. What to build now (summary)

A crisp recap of the base system to deliver:

1. Counterparty records (clients, suppliers) with three-letter codes, managed by salesperson and purchasing behind a visibility wall.
2. A product catalogue with the specification fields; standard products carry agreed prices.
3. Deals: bilateral, a directional code (SELLER-BUYER-NNN), an order (spec plus terms), a direction (sell or buy), a stage, and a document set.
4. The spine: one traceable identity per batch, spec to lot, child IDs for splits, a rolled-up status; deals reference it; the system stamps it when spawning the buy side.
5. The five-stage lifecycle plus cancelled, per-deal status and spine rollup, the configurable gate engine, and chain-break flagging on cancel.
6. Documents on the deal (quotation to order specification, proforma, contract, purchase order, packing list, CMR, invoices), plus the one-button transport pack.
7. The group-and-rights access model: deal-level isolation plus field-level visibility, granular editable rights, default groups, all configurable in the application.
8. AI document handling: inbound ingestion, document generation, and outbound to logistics with AI-read replies.
9. Configurability throughout: groups, rights, gates, products and counterparties all editable in settings, not in code.
10. Activities by stage and direction are documented (Section 7) for understanding, but are not built as enforced tasks in this version.

12. Visual identity (brand palette)

The platform interface uses the Timber World palette. These are the canonical colour values for the build; this document is styled with the same palette.

12.1 Brand colours

Colour	Name	Role
#1B4332	Forest Green	Primary brand colour, main actions and links.
#8B5A2B	Warm Oak	Secondary accent, the wood theme.
#FAF6F1	Warm Cream	Background, warm neutral.
#2D3436	Charcoal	Text and dark elements.

12.2 Status colours (badges)

Status badges across the interface use the colours below. The deal-stage badges map naturally onto them — Draft to the Draft colour, Delivered to Success, and Cancelled to Error — as the lifecycle diagram in Section 6 reflects.

Status	Badge colour	Use
Draft	#D89B33	Draft or unsaved state.
Pending	#F6D44B	Awaiting action or confirmation.
Success	#2E9748	Completed or confirmed.
Error	#CA3733	Error or failure.
Info	#2682CC	Informational.
Warning	#F6A338	Needs attention.

In short: Forest Green for primary actions and headers, Warm Oak for secondary accents and the wood theme, Warm Cream as the light background, and Charcoal for text. Status colours indicated above are approximate and may be tuned to meet contrast requirements.

Glossary

Term	Meaning
Deal	A transaction between two parties, one seller and one buyer; self-contained, with its own price, terms and documents.
Chain	A series of linked deals through which goods pass from producer to final customer.
Spine	The traceable identity of a batch of goods running through the chain; the one thread that links the deals. Has a spec (planned) phase and a lot (physical) phase.
Spec / specification	The product definition (species, type, finish, certificate, dimensions, pieces, volume); also the planned state of the spine.
Order	A deal's content: the product definition plus that deal's commercial terms.
Lot / package	The spine once the goods physically exist.
Physical shipment	A movement of goods (a truck); may underlie one or several deals.
Sell deal	A deal in which the operating company is the seller (faces the customer).
Buy deal	A deal in which the operating company is the buyer (faces the supplier).
Seller / buyer	The two parties of a single deal; fixed by that deal.
Counterparty	A company record (client or supplier) referenced by deals.
Group	An editable set of rights; users inherit their access from their group.
Gate	The set of confirmations required to move a deal from one stage to the next; configurable and switchable.
Super administrator	The owner; sees the whole chain and configures the system.